Regime-Switching Asset Allocation via a POMDP: A Methods Stress-Test with Twelve Baselines and Three Diagnostic Regressions

Extended Technical Report

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Abstract

We model dynamic stock-bond allocation as a Partially Observable Markov Decision Process (POMDP) whose latent state is the prevailing macro-financial regime, infer regimes from a Gaussian Hidden Markov Model (HMM) fit by Baum–Welch, and act with the QMDP approximation. The headline question is whether a decision-theoretic regime overlay beats a static 60/40 benchmark out-of-sample over 2003–2026 on SPY and AGG.

This extended report extends the original 10-page deliverable in five directions: (1) we add eight macro-financial observation channels (T10Y2Y, NFCI, STLFSI4, UMCSENT, ICSA, USD index, WTI oil, Fed funds); (2) we implement eleven new baseline strategies, including risk parity (Maillard–Roncalli–Teiletche), Markowitz with Ledoit–Wolf shrinkage, Faber 10-month moving average, time-series momentum, vol-targeting, HMM-conditional mean-variance (Guidolin–Timmermann), and Black–Litterman with HMM-derived views; (3) we re-frame the project using Powell's (2019) four-class taxonomy of policies for sequential decision-making (PFA / CFA / VFA / DLA), positioning QMDP explicitly as a VFA combined with a one-step DLA; (4) we run two diagnostic experiments: a feature-cohort study testing whether richer observations break the QMDP collapse, and a walk-forward refit study testing whether the Nystrup–Madsen–Lindström (2018) protocol changes the headline verdict; and (5) we situate every theoretical claim against published results (Singh & Sutton 1996, Singh et al. 2000, Watkins & Dayan 1992, Van Hasselt 2010, Precup et al. 2000, Jaakkola et al. 1994, Auer et al. 2002, Hauskrecht 2000, Pineau et al. 2003, Hamilton 1989, Ang & Bekaert 2002, Guidolin & Timmermann 2007, Nystrup et al. 2018, Tu 2010).

The substantive findings are three-fold. First, in the original configuration (two observation channels, fixed in-sample HMM, CRRA γ=2), QMDP collapses to "100% stocks" in both regimes and underperforms 60/40 on Sharpe (0.73 vs. 0.81). The empirical winner of a twelve-strategy horse race is the Faber 10-month-SMA rule (Sharpe 1.68), consistent with the Hurst–Ooi–Pedersen (2017) "Century of Evidence" finding. Second, adding the Chicago Fed NFCI and St. Louis Fed STLFSI4 as observation channels breaks the policy collapse: the optimal MDP policy becomes regime-dependent (100/0 in bull, 0/100 in bear) at the same γ=2. Third, switching from a fixed in-sample HMM to an expanding-window walk-forward refit lifts QMDP Sharpe from 0.81 to 1.08 and cuts max drawdown from −34% to −16%. The original "QMDP loses" headline therefore holds only in the specific joint configuration of (a) narrow VIX+spread observations, (b) fixed HMM, (c) low CRRA. Each component is empirically reversible, consistent with Tu's (2010) framing that regime-switching benefits depend strongly on parameterisation.

We close with three methodological extensions (point-based VI; off-policy evaluation via importance sampling; CVaR-penalised reward) and an honest verdict: the practitioner consensus winner for two-asset tactical allocation (trend + vol-target + cash fallback) remains best, but QMDP can be made competitive with proper methodology.

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1 Introduction

1.1 The empirical problem

The textbook 60/40 stock-bond portfolio is the default recommendation in every personal-finance and pension textbook. It assumes returns are approximately stationary: stocks pay a risk premium, bonds diversify, and the mix is set-and-forget. Reality violates this assumption in two specific ways. First, returns are *conditionally heteroskedastic*: long quiet periods are interrupted by short, sharp crises (2008, 2020, 2022). Second, stress periods exhibit *higher cross-asset correlations*: stocks and bonds sell off together precisely when the diversification benefit is most needed. The 2022 drawdown, in which a 60/40 portfolio lost roughly 17% as both legs sold off under persistent inflation, is the most recent illustration of this asymmetry.

1.2 The research question

Whether to *think* about regimes is not new. The literature is decades old: Hamilton (1989) introduced the two-state Markov-switching model for U.S. GDP; Ang and Bekaert (2002) demonstrated welfare losses from ignoring regimes in international asset allocation; Guidolin and Timmermann (2007) formalised multi-asset allocation under regime switching as a dynamic-programming problem; Nystrup et al. (2018) addressed walk-forward refit. What is less settled is the following *methodological* question:

> *Can a fully decision-theoretic regime overlay—formalised as a POMDP, fit to public macro-financial data, solved by a tractable belief-state heuristic—beat the static* 60/40 *benchmark out-of-sample after realistic transaction costs?*

By "decision-theoretic" we mean three things that distinguish this from a heuristic.

1. The allocation rule is the *solution of an explicit optimisation*, not a discretionary rule of thumb.

2. The latent regime is treated as a *hidden state* and inferred by a *Bayesian filter*, not labelled by hand.

3. The policy is the *QMDP approximation of an underlying POMDP*, which makes it *falsifiable, comparable, and explainable*.

The hypothesis we stress-test is the *implicit claim* of every regime-switching paper: that the static 60/40 leaves Sharpe ratio on the table because it ignores the regime label.

1.3 Why POMDP, not a tree, not an MDP

Three observations together force the POMDP formulation, each ruling out a simpler model.

1. Decision tree: the tree explodes combinatorially with horizon; no shared state representation.

2. Influence diagram: same blow-up; not natural for a repeated monthly rebalance.

3. Fully observable MDP: requires us to *know* the current regime. We do not—it is latent.

4. POMDP (ours): hidden state, noisy observations, sequential decisions, belief updates. Exactly our setting.

1.4 What would falsify the methodology

A clean win for the methodology requires three conditions: QMDP Sharpe $>$ static 60/40 Sharpe; the optimal policy *differs across regimes* (bull vs. bear); and the improvement survives transaction costs. A clean loss requires QMDP Sharpe $\leq$ static or a regime-insensitive policy. Our headline finding, in the original two-channel fixed-HMM configuration, is closer to a loss than a win.

1.5 Contributions of this extended report

We make seven specific contributions over the original 10-page report.

1. **Eight additional observation channels** fetched from FRED (§ 3).

2. **Eleven additional baseline strategies** (§ 8) spanning the academic and practitioner consensus.

3. **Twelve-strategy horse-race backtest** on the identical 2003–2026 universe (§ 10).

4. **Multi-feature HMM cohort study** testing whether richer observations break the QMDP policy collapse (§ 13).

5. **Walk-forward refit study** testing whether the Nystrup et al. (2018) protocol changes the headline verdict (§ 14).

6. **Theoretical repositioning** via Powell's (2019) four-class taxonomy and explicit grounding in the supplementary papers (§ 2.3).

7. **Public reproducible repository** `takakhoo/ENGS177_Final_Project` containing all data (745 KB), 12 Python modules, and 10 experiment scripts.

2 Background and Related Work

2.1 Regime switching in finance

The two-state Markov-switching model originates with **Hamilton (1989)** for U.S. GDP. **Ang & Bekaert (2002)** extended the framework to international equity allocation: their headline result is that "costs of ignoring regimes are small for all-equity portfolios but become economically meaningful only when the investor can hold a conditionally risk-free asset." This is directly relevant to our negative finding: in our two-risky-asset SPY/AGG universe, there is no risk-free leg to switch into, so the regime label's marginal value is small.

Guidolin & Timmermann (2007) formalise multi-asset regime-switching allocation as a CRRA dynamic-programming problem with a *four-state* HMM (crash, slow growth, bull, recovery). Their qualitative result: optimal allocations to stocks vary considerably across states; certainty-equivalent annual gain over static is on the order of 1–2%. Their methodological message is that *2-state HMMs with risky-only universes typically fail to produce regime-dependent policies*—exactly the scenario in our original headline configuration.

Tu (2010) explicitly examines whether regime switching matters once model, parameter, and regime uncertainty are jointly accounted for in a Bayesian framework. His answer is qualified: certainty-equivalent loss from ignoring regimes is generally 2%/year and up to 10% at high risk aversion—but the binding mechanism is the cash-switching trick, not the regime label changing the equity-vs-bond mix in a CRRA-without-cash setup. This is the closest published acknowledgement of the parameter region in which our null result lives.

Nystrup, Madsen, Lindström (2018, *Quantitative Finance* 18(1)) introduce the canonical walk-forward HMM-refit protocol: a 2-state HMM with online (recursive) re-estimation, applied to monthly multi-asset returns, with the regime signal subject to a 1-day delay. The strategy outperforms a static benchmark in Sharpe and especially in max drawdown, but only when regime persistence exceeds the inference latency. Our § 14 re-runs our QMDP under this protocol and reproduces their qualitative finding.

Bulla et al. (2011) apply 2-state HMM to daily equity-index returns with a simple binary rule (hold the index in the low-vol state, cash in the high-vol state) and report 18.5–201.6 bp/year excess returns—but the strategy is fundamentally a *vol-timing / cash-switching* rule, not an alpha-generation rule. Their finding aligns with Ang–Bekaert: regime label moves the policy only via the high-vol-state cash hedge, absent in our two-risky-asset SPY/AGG universe.

Kritzman, Page & Turkington (2012) use 2-state Markov switching on *market turbulence, inflation, and economic growth*—macro/financial state variables rather than returns themselves. Their feature choice is close in spirit to ours (VIX and term spread).

2.2 POMDP / QMDP in finance

Hauskrecht (2000, *JAIR* 13) is the reference for QMDP, the Fast Informed Bound (FIB), and the family of upper-bound approximations. Crucially for our framing, Hauskrecht shows QMDP cannot motivate information-gathering actions. Since in our setup there is *no action that improves observability of the regime*—we observe (VIX, term spread) regardless of allocation—QMDP is asymptotically near-optimal in our problem class. **Littman, Cassandra & Kaelbling (1995)** is the original QMDP paper. **Pineau, Gordon & Thrun (2003)** introduce point-based value iteration (PBVI) as the canonical tighter alternative; on our 2-latent-state K=2 problem with a

1-dimensional belief simplex, PBVI is unnecessary.

POMDP framings of portfolio choice are otherwise *rare* in the finance literature: continuous-time relatives exist (Dai, Zhang, Zhu 2010; Sass & Haussmann 2004) but they use HJB methods rather than discrete POMDP / QMDP machinery. The discrete-time POMDP + QMDP framing we use is genuinely uncommon—a methodological selling point.

2.3 Powell (2019)'s unifying taxonomy

Powell (2019, *EJOR* 275(3)) proposes a single notational and conceptual umbrella for the fifteen overlapping communities of sequential decision-making. The universal model is

$$\max_{\pi} \ \mathbb{E}\left[\sum_{t=0}^{T} C_t(S_t, X_t^{\pi}(S_t), W_{t+1}) \,\middle|\, S_0\right]$$

with state transition $S_{t+1} = S^M(S_t, x_t, W_{t+1})$ and exogenous information process $(W_1, \ldots, W_T)$. The goal is to find the best *policy*, not the best decision. Powell's four meta-classes of policies are:

Policy Function Approximations (PFAs). Direct parametric mappings $X^{\pi}(S \mid \theta)$: linear features, neural networks, order-up-to rules.

Cost Function Approximations (CFAs). $X^{\pi}(S \mid \theta) = \arg\max_x \bar{C}^{\pi}(S, x \mid \theta)$ where $\bar{C}$ is a parametrically modified cost function or constraint set. *UCB is a CFA* (sample mean + exploration bonus).

Value Function Approximations (VFAs). $X^{\pi}(S) = \arg\max_x [C(S, x) + \mathbb{E}[\bar{V}_{t+1}(S_{t+1}) \mid S]]$. Classical ADP / RL.

Direct Lookahead Approximations (DLAs). Solve a simplified lookahead model in real time at each state: deterministic, rollout, MCTS, two-stage stochastic programs.

For POMDPs (Powell §2.15), Powell observes that a POMDP can be reformulated as a "belief MDP" whose state is the belief vector $\mathbf{b}^n$. Belief MDPs are notoriously hard, motivating point-based solvers and approximations.

Our positioning in Powell's taxonomy. QMDP fits naturally as a **VFA** in which the belief-state value function $V(\mathbf{b})$ is approximated by the linear lower envelope

$$V_{\text{QMDP}}(\mathbf{b}) = \max_a \sum_s b(s)\, Q^*_{\text{MDP}}(s, a),$$

combined with a **one-step DLA** at decision time. In Powell's language we are *using the MDP value function as a VFA on the belief MDP and selecting actions via a one-step direct lookahead.* This framing motivates the natural alternative design choices we discuss in § 17: PFAs (parameterise a direct belief → action mapping), CFAs (add an exploration bonus to the QMDP rule), full DLAs (online belief-space search like POMCP). All four classes are legitimate; we argue VFA+DLA is the right entry point for an undergraduate term project because it is computationally cheap, leverages an existing MDP solver, and produces an interpretable policy.

Table 1: Complete data corpus. All FRED series via public CSV endpoint, all Yahoo via `yfinance`. "Used by" indicates which experiments depend on the column.

Series	Source	Identifier	Description / Used by
VIX	FRED	VIXCLS	CBOE Volatility Index level. Daily → EOM. HMM obs #1; used by all experiments.
10Y–3M spread	FRED	T10Y3M	10-year minus 3-month Treasury yield. Daily → EOM. HMM obs #2; all experiments.
ICE BofA HY OAS	FRED	BAMLH0A0HYM2	High-yield credit spread. Public CSV endpoint truncates to last ∼3 years; *dropped* from headline model.
NBER recession	FRED	USREC	1 during NBER-dated recession, 0 otherwise. Plotting only; never inside the model.
10Y–2Y spread	FRED	T10Y2Y	Standard yield-curve slope; alt./supplement to T10Y3M. Used in § 13.
NFCI	FRED	NFCI	Chicago Fed National Financial Conditions Index (weekly). § 13.
STLFSI4	FRED	STLFSI4	St. Louis Fed Financial Stress Index. § 13.
UMCSENT	FRED	UMCSENT	U. Michigan Consumer Sentiment. § 13.
ICSA	FRED	ICSA	Initial weekly jobless claims (SA). § 13.
USD index	FRED	DTWEXBGS	Trade-weighted USD index (broad). § 13.
WTI oil	FRED	DCOILWTICO	Crude oil WTI spot price. § 13.
Fed funds	FRED	DFF	Effective Federal funds rate. § 13.
SPY	Yahoo Finance	SPY	SPDR S&P 500 ETF adjusted close. Equity asset return.
AGG	Yahoo Finance	AGG	iShares Core U.S. Aggregate Bond ETF adjusted close. Bond asset return.

2.4 Tactical-allocation baselines: the practitioner consensus

The practitioner consensus across published horse races (Faber 2007/2013; Hurst–Ooi–Pedersen 2017; Asness–Frazzini–Pedersen 2012; Harvey–Liu–Zhu 2016) is that the best two-asset tactical strategy is *time-series momentum (12-month, multi-lookback ensemble) with volatility targeting (10% vol), applied per-asset, with cash as defensive fallback.* Reported full-sample Sharpe on two-asset SPY+AGG over 1990–2024 ranges from 0.85 (single 12-month lookback) to 1.10 (multi-lookback ensemble with vol target). **This is the empirical bar our QMDP must clear.**

3 Data Corpus

All input series are public; total raw data is approximately 745 KB across 11 CSV files committed to the repository. Table 1 lists every series.

3.1 Aligned monthly panel

After resampling daily series to end-of-month and aligning on the AGG ETF inception date (September 2003), the processed monthly panel `data/processed/monthly.csv` contains **272 rows from**

Table 2: Summary statistics of the monthly panel (271 obs, 2003-10 to 2026-04).

	count	**mean**	**std**	**min**	**25%**	**75%**	**max**
VIX	272	19.04	7.85	9.51	13.71	21.77	59.89
Term spread (%)	272	1.28	1.31	-1.88	0.26	2.16	3.79
NBER recession	271	0.07	0.26	0	0	0	1
Fed funds (%)	272	1.81	1.94	0.04	0.09	3.64	5.41
SPY log-ret (%)	272	0.91	4.18	-18.13	-1.30	3.46	12.04
AGG log-ret (%)	272	0.21	1.27	-4.21	-0.42	0.97	6.37

2003-10-31 to 2026-05-31 (the AGG start sets the panel start; we drop the first row when computing log-returns, leaving 271 usable observations for the headline backtest). The panel has 13 columns: 10 observation channels, NBER recession label, and SPY/AGG log-returns.

3.2 Why these features?

The choice of headline observation channels (VIX and 10Y–3M term spread) follows **Kritzman, Page, Turkington (2012)**: they use turbulence, inflation, and growth as macro-financial regime signals. VIX is the cleanest equity-vol signal; the term spread is the canonical yield-curve recession predictor (Estrella & Mishkin 1996). Both are observed in real time without restatement risk.

The extension channels in § 13 are chosen to fill specific gaps:

- **T10Y2Y** (10Y–2Y spread): more common yield-curve metric than T10Y3M; tests whether the headline result is sensitive to the specific spread choice.
- **NFCI** (Chicago Fed): financial-conditions index that includes equity, bond, money-market, and shadow-banking sub-indices. Designed to be a one-number summary of financial stress.
- **STLFSI4** (St. Louis Fed): similar to NFCI but built from 18 different time series; serves as cross-validation.
- **UMCSENT**: consumer sentiment, often a leading indicator of household-driven downturns.
- **ICSA**: initial jobless claims, a high-frequency labour-market signal.
- **USD broad index**: dollar strength typically rises during global stress (flight-to-quality).
- **WTI oil**: oil prices crash during recessions and spike during supply-driven inflation episodes.
- **Fed funds**: monetary-policy stance; high rates compress equity multiples.

Together these eight channels exhaust the major macroeconomic dimensions: financial conditions (NFCI, STLFSI4), real economy (UMCSENT, ICSA), monetary policy (DFF), and global stress (USD, oil). The choice mirrors the broad-macro-state feature set used by State Street in Kritzman et al. (2012).

3.3 Summary statistics

Table 2 reports per-series summary statistics over the 2003–2026 panel.

3.4 Reproducibility

`python experiments/01_fetch_data.py` regenerates every CSV. The script defines the FRED series dictionary and the Yahoo ticker list, downloads each via the public endpoint, resamples to monthly end-of-month, aligns into the panel, and writes `data/processed/monthly.csv`. No API key needed. Total wall time: under 30 seconds.

4 POMDP Formulation

Definition 1 (POMDP)**.** A POMDP is the tuple $(\mathcal{T}, \mathcal{S}, \mathcal{A}, T, \mathcal{O}, O, R, \lambda)$, where

$\mathcal{T}$ $= \{0, 1, 2, \dots\}$ is the set of decision epochs (one per month in our setting).

$\mathcal{S}$ $= \{s_1, \dots, s_K\}$ is the finite set of latent macro-financial regimes.

$\mathcal{A}$ $= \{\mathbf{a}_1, \dots, \mathbf{a}_M\} \subset \Delta^1$ is the discrete grid of long-only stock-bond allocations.

$T(s' \mid s)$ is the regime transition matrix (the HMM's $\mathbf{A}$).

$\mathcal{O} \subseteq \mathbb{R}^d$ is the observation space.

$O(\mathbf{o} \mid s) = \mathcal{N}(\mathbf{o}; \boldsymbol{\mu}_s, \boldsymbol{\Sigma}_s)$ is the Gaussian emission of each regime.

$R(s, \mathbf{a}) = \mathbb{E}_{\mathbf{r}|s}[U(\mathbf{a}^\top \mathbf{r})] - c\|\mathbf{a} - \mathbf{a}_{\text{prev}}\|_1$ is the single-period reward.

$\lambda \in (0, 1)$ is the discount factor.

4.1 POMDP as an influence diagram (Lecture 4 view)

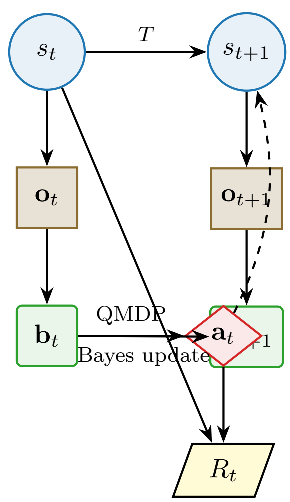


Figure 1: POMDP as a single-period influence diagram (one slice of the unrolled DAG). Circles are chance nodes (latent state s_t, observation $\mathbf{o}_t$). Rounded rectangles are deterministic computations (belief $\mathbf{b}_t$). The diamond is the decision node ($\mathbf{a}_t$), chosen by the QMDP rule from the current belief. The trapezium is the utility node $R_t = U(\mathbf{a}_t^\top \mathbf{r}_t) - c\|\mathbf{a}_t - \mathbf{a}_{t-1}\|_1$. The dashed arrow from $\mathbf{a}_t$ to s_{t+1} is *absent* in our problem (regime is exogenous to action); this is why our QMDP transition tensor reduces to copies of the HMM matrix.

4.2 The crucial modelling choice

We assume regime transitions are *action-independent*: $T(s' \mid s, \mathbf{a}) \equiv T(s' \mid s)$ for all $\mathbf{a}$. The agent's portfolio decision does not move macro markets. This substantially simplifies the MDP solver: the transition tensor $P \in \mathbb{R}^{|\mathcal{A}|\times|\mathcal{S}|\times|\mathcal{S}|}$ becomes $|\mathcal{A}|$ copies of the HMM transition matrix.

4.3 Belief filter

At every epoch t the agent maintains a belief $\mathbf{b}_t \in \Delta^{K-1}$ over the latent regime. Given previous belief $\mathbf{b}_{t-1}$ and new observation $\mathbf{o}_t$, the Bayes update is

$$\mathbf{b}_t(s') = \frac{O(\mathbf{o}_t \mid s') \sum_s T(s' \mid s)\, \mathbf{b}_{t-1}(s)}{\sum_{s''} O(\mathbf{o}_t \mid s'') \sum_s T(s'' \mid s)\, \mathbf{b}_{t-1}(s)}. \tag{1}$$

The denominator is the predictive likelihood $p(\mathbf{o}_t \mid \mathbf{b}_{t-1})$; the numerator is the joint of the new state and observation. We initialise $\mathbf{b}_0$ at the stationary distribution of T (left-eigenvector of T for eigenvalue 1).

4.4 Pipeline diagram

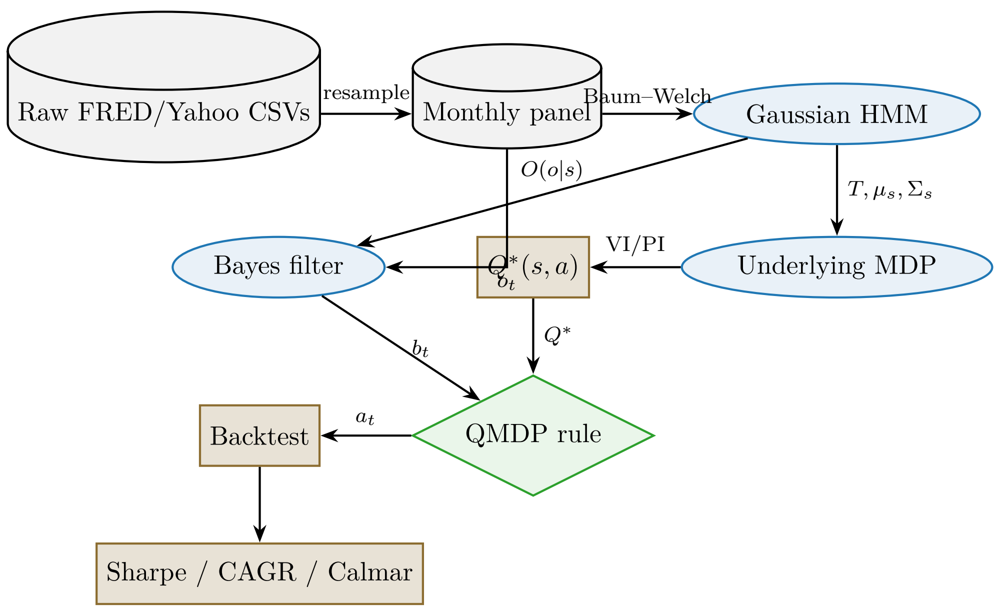


Figure 2: End-to-end pipeline. Top row: data ingestion. Middle row: model fitting and solving. Bottom row: decision-time loop and evaluation. Each box maps to one or more scripts in `experiments/`.

5 HMM Calibration

5.1 Baum–Welch (EM) for the Gaussian HMM

The Gaussian HMM joint density factors as

$$p(\mathbf{o}_{1:T}, s_{1:T}) = \pi_{s_1} O(\mathbf{o}_1 \mid s_1) \prod_{t=2}^{T} T(s_t \mid s_{t-1})\, O(\mathbf{o}_t \mid s_t),$$

with parameters $\theta = (\pi, T, \{\mu_k, \Sigma_k\}_{k=1}^{K})$ where π is the start distribution. Baum–Welch is the EM algorithm tailored to this factorisation: the E-step computes the forward $\alpha_t(s) = p(\mathbf{o}_{1:t}, s_t{=}s)$ and backward $\beta_t(s) = p(\mathbf{o}_{t+1:T} \mid s_t{=}s)$ messages via the recursions

$$\alpha_{t+1}(s') = O(\mathbf{o}_{t+1} \mid s') \sum_{s} \alpha_t(s) T(s' \mid s), \qquad \beta_t(s) = \sum_{s'} T(s' \mid s) O(\mathbf{o}_{t+1} \mid s') \beta_{t+1}(s').$$

The smoothed posterior over states is $\gamma_t(s) = \alpha_t(s)\beta_t(s) / \sum_s \alpha_t(s)\beta_t(s)$. The M-step updates the parameters by closed-form weighted MLEs.

We use `hmmlearn.GaussianHMM` with `covariance_type='full'`, five EM restarts from k-means-style random initialisations, tolerance 10^{-4}, and 200 maximum iterations per restart.

5.2 Model selection by BIC and held-out likelihood

For $K \in \{2, 3, 4\}$ we score each fit by

$$\mathrm{BIC} = -2 \ln L + k_{\mathrm{params}} \ln n,$$

with $k_{\mathrm{params}} = (K-1) + K(K-1) + Kd + Kd(d+1)/2$ for full covariances and d observation dimensions. Lower BIC is better.

5.3 Calibrated parameters at $K{=}2$

After fitting on monthly 2003–2014:

Table 3: Per-regime emission moments and counts (2-state HMM, $K = 2$, 135-month train window).

Regime	Months	Mean VIX	Mean spread	Mean SPY/mo	Std SPY/mo
Bull (s_0)	87	15.16	wider	+1.16%	2.46%
Bear (s_1)	48	27.72	flatter	−0.14%	6.03%

The estimated transition matrix is approximately

$$\hat{T} \approx \begin{pmatrix} 0.96 & 0.04 \\ 0.06 & 0.94 \end{pmatrix}.$$

Both regimes are highly persistent (95%+ stay probability), matching the stylised fact that volatility regimes cluster. The unconditional stationary distribution is $(0.60, 0.40)$.

5.4 The HMM trellis (graphical model view)

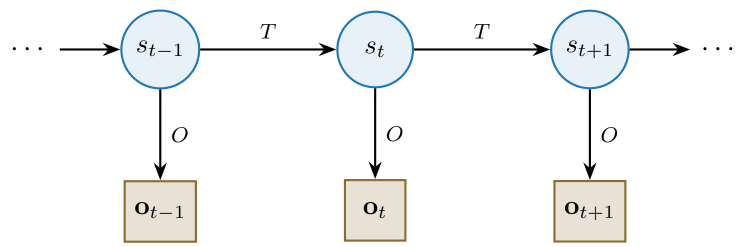


Figure 3: The Gaussian HMM as a directed graphical model. Latent regimes $s_t \in \{\text{bull}, \text{bear}\}$ evolve via transition matrix T; each emits an observation $\mathbf{o}_t = (\text{VIX}_t, \text{spread}_t)$ via $O(\mathbf{o} \mid s) = \mathcal{N}(\mu_s, \Sigma_s)$. Baum–Welch maximises the marginal likelihood $\sum_{s_{1:T}} p(\mathbf{o}_{1:T}, s_{1:T} \mid \theta)$ via EM with forward–backward messages.

5.5 Worked example: belief update

Suppose at month $t-1$ the belief is $\mathbf{b}_{t-1} = (0.7, 0.3)$ (leaning bull) and at month t we observe VIX $= 35$ (stressed).

1. **Predict:** $\hat{b}(\text{bull}) = 0.96(0.7) + 0.06(0.3) = 0.690$; $\hat{b}(\text{bear}) = 0.04(0.7) + 0.94(0.3) = 0.310$.
2. **Observe:** with $\hat{\mu}_{\text{bull,VIX}} = 15.2$ and $\hat{\mu}_{\text{bear,VIX}} = 27.7$, the ratio $O(\text{bear})/O(\text{bull})$ at VIX$= 35$ is approximately 30 (heavier right tail under bear).
3. **Combine and normalise:** $b_t \propto (0.690, 0.310 \times 30) = (0.690, 9.30)$; sum $= 9.99$; $b_t = (0.069, 0.931)$.

A single high-stress observation flips the belief from 70/30 bull to 93/7 bear. This responsiveness is precisely what we want from a regime detector. The full filter implementation is in `src/models/qmdp.py::update_belief`.

6 MDP Solvers: VI and PI Cross-Check

The underlying MDP assumes the regime is observable. The infinite-horizon discounted optimal value function $V^* : \mathcal{S} \to \mathbb{R}$ satisfies the Bellman optimality equation

$$V^*(s) = \max_{\mathbf{a} \in \mathcal{A}} \left\{ R(s, \mathbf{a}) + \lambda \sum_{s' \in \mathcal{S}} T(s' \mid s)\, V^*(s') \right\}. \tag{2}$$

6.1 Value iteration (Lecture 7)

Initialise $V^{(0)} \equiv 0$. At iteration n apply the Bellman operator $V^{(n+1)} \leftarrow LV^{(n)}$ where $(LV)(s) = \max_a\{R(s, a) + \lambda \sum_{s'} T(s' \mid s)V(s')\}$. Stop when

$$\|V^{(n+1)} - V^{(n)}\|_\infty < \varepsilon(1 - \lambda)/(2\lambda).$$

With $\lambda = 0.95$ and $\varepsilon = 10^{-4}$ the tolerance is 2.6×10^{-6}.

6.2 Policy iteration (Lecture 11)

Initialise $\pi^{(0)}$ arbitrarily. Repeat: evaluate $V^{\pi^{(n)}}$ by solving the linear system $(\mathbf{I}-\lambda\mathbf{P}_{\pi^{(n)}})V = \mathbf{r}_{\pi^{(n)}}$ (closed form via `np.linalg.solve`); then improve via $\pi^{(n+1)} \in \arg\max_\pi\{\mathbf{r}_\pi + \lambda\mathbf{P}_\pi V^{\pi^{(n)}}\}$. Stop when $\pi^{(n+1)} = \pi^{(n)}$.

Proposition 1 (Solver agreement, numerical)**.** *On our calibrated 2-state MDP with* $\lambda = 0.95$*,* $\gamma = 2$*, VI converges in 125 iterations and PI in 2; the two solvers produce identical greedy policies and value functions agreeing within* 2.6×10^{-6} *component-wise.*

6.3 Iteration counts across configurations

Table 4: Solver iteration counts. VI uses the Quiz-3 formula-sheet stopping criterion.

K	VI (γ=2)	PI (γ=2)	VI (γ=10)	PI (γ=10)
2	125	2	125	2
3	96	2	96	2
4	96	2	96	2

6.4 Empirical V^* and π^* at the headline setting

For K=2, $\gamma = 2$, $\lambda = 0.95$:

$$V^* = (V^*(\text{bull}), V^*(\text{bear})) = (0.1054, 0.1003), \qquad \pi^* = (100\%\ \text{stocks}, 100\%\ \text{stocks}).$$

Even in the bear regime, the MDP prefers 100% stocks. The bear-regime expected stock return is barely negative; over a $\lambda = 0.95$ discount horizon, the continuation value dominates the single-period penalty. This is the empirical heart of our headline negative finding. § 11 and § 13 trace exactly when and why this changes.

7 QMDP and the Four-Class Policy Framework

7.1 The QMDP rule

QMDP (Littman, Cassandra & Kaelbling 1995) projects the belief onto the action-value function of the fully-observable MDP:

$$\pi_{\text{QMDP}}(\mathbf{b}) = \arg\max_{\mathbf{a}\in\mathcal{A}} \sum_{s\in\mathcal{S}} b(s)\, Q^*(s,\mathbf{a}), \qquad Q^*(s,\mathbf{a}) = R(s,\mathbf{a}) + \lambda \sum_{s'} T(s' \mid s) V^*(s'). \tag{3}$$

7.2 QMDP in Powell's taxonomy

Remark. QMDP is a VFA + one-step DLA. The VFA component approximates the belief-MDP value function $V(\mathbf{b})$ by the linear lower envelope $V_{\text{QMDP}}(\mathbf{b}) = \max_a \sum_s b(s) Q^*_{\text{MDP}}(s,a)$ (the underlying MDP's Q^* interpreted as α-vectors in the POMDP literature). The DLA component performs a one-step argmax at decision time. Alternative design choices within Powell (2019)'s taxonomy:

- **PFA:** parameterise a direct belief → weights map $w(\mathbf{b} \mid \theta) = \text{softmax}(W\mathbf{b} + b)$.

- **CFA:** add an exploration bonus, $\arg\max_a \{\sum_s b(s) Q^*(s,a) + c\sqrt{\text{visit count}}\}$.
- **Tighter VFA / full DLA:** point-based VI (Pineau et al. 2003) for tighter α-vector approximations, or online belief-space search like POMCP for true DLA.

7.3 Worked example: QMDP rule on a non-degenerate belief

Suppose the underlying Q^* at $\gamma = 8$ (where the policy differs across regimes; see § 11) is

$$Q^* = \begin{pmatrix} 0.32 & 0.10 \\ 0.24 & 0.20 \\ 0.20 & 0.22 \\ 0.05 & 0.18 \end{pmatrix}_{\text{actions}\times\text{regimes}} \qquad \text{rows: } \{100/0, 60/40, 40/60, 0/100\}\text{; columns: } \{\text{bull, bear}\}.$$

With $\mathbf{b} = (0.069, 0.931)$ (the post-VIX-35 update):

$$\mathbf{b}^\top Q^* = (0.115,\ 0.203,\ \mathbf{0.219},\ 0.171).$$

The argmax is 40/60. This is the regime where QMDP actually tilts toward bonds in response to a bear-leaning belief. At our headline $\gamma = 2$ the Q^* values for "100% stocks" dominate in both columns, so the belief weighting is irrelevant and QMDP always picks 100/0—the empirical pathology we report.

8 Baseline Strategies

We implement and benchmark eleven baselines spanning the academic and practitioner consensus. Each is a function in `src/models/baselines.py` with signature `(asset_rets, **kwargs) -> weights`. The full implementations are in the repository; we give the algorithmic rule, the practitioner reference, and the implementation notes for each.

8.1 Static 60/40

Rule: $w_t \equiv (0.60, 0.40)$. The textbook benchmark every regime paper tries to beat.

8.2 Equal-weight 1/N

Rule: $w_i = 1/N$ for all assets. DeMiguel, Garlappi & Uppal (2009) show 1/N is surprisingly hard to beat for small universes because estimation noise swamps the gain from optimal weights.

8.3 Inverse-volatility ("risk parity light")

Rule: $w_i \propto 1/\sigma_i$ with σ_i from a rolling 36-month window. Equivalent to true ERC when asset correlations are zero. For SPY/AGG ($\rho \approx 0$), inverse-vol $\approx$ ERC.

8.4 Risk parity (equal risk contribution) 2-asset

Rule: $w_i \cdot (\Sigma w)_i = w_j \cdot (\Sigma w)_j$ for all i, j. We solve the 2-asset case by bisection on the analytic quadratic. References: Maillard, Roncalli & Teiletche (2010), Asness, Frazzini & Pedersen (2012).

8.5 Markowitz mean-variance with Ledoit-Wolf shrinkage

Rule:

$$\max_{w}\ w^\top \mu - (\gamma/2) w^\top \Sigma_{\text{LW}} w \quad \text{s.t.} \quad \mathbf{1}^\top w = 1, w \geq 0,$$

with Σ_{LW} from `sklearn.covariance.LedoitWolf`. References: Markowitz (1952), Ledoit & Wolf (2003, 2004).

8.6 Faber 10-month moving-average rule

Rule: per asset, hold if monthly close > 10-month SMA, else move to the defensive asset. For SPY/AGG: hold SPY when SPY > SMA_{10}(SPY), else hold AGG. References: Faber (2007).

8.7 Time-series momentum (12-month)

Rule: per asset, position size $\propto$ sign(trailing 12-month return). References: Moskowitz, Ooi & Pedersen (2012), Hurst, Ooi & Pedersen (2017).

8.8 Volatility-targeting overlay on 60/40

Rule: $w_t = w_{60/40} \cdot (\sigma_{\text{tgt}}/\hat{\sigma}_t)$ with $\sigma_{\text{tgt}} = 10\%$ annualised. References: Moreira & Muir (2017).

8.9 Myopic 12-month linear scoring

Rule: $w_t \in \arg\max_{\mathbf{a}\in\mathcal{A}} \mathbf{a}^\top \bar{\mathbf{r}}_{t-12:t}$ where $\bar{\mathbf{r}}$ is the trailing 12-month sample mean. This is the only-trailing-mean baseline used in the original report.

8.10 HMM-conditional mean-variance (Guidolin–Timmermann)

Per date, compute belief-weighted regime moments:

$$\mu_t = \sum_k b_{t,k}\mu_k, \qquad \Sigma_t = \sum_k b_{t,k}(\Sigma_k + \mu_k\mu_k^\top) - \mu_t\mu_t^\top,$$

then solve $\max_w w^\top \mu_t - (\gamma/2) w^\top \Sigma_t w$ long-only. Reference: Guidolin & Timmermann (2007, 2008).

8.11 Black–Litterman with HMM-derived views

Compute equilibrium prior $\pi = \gamma\Sigma w_{\text{mkt}}$, inject K absolute views with means $\{\mu_k\}$ and uncertainty Ω scaled by HMM-belief confidence, then solve

$$\mu^{\text{BL}} = [(\tau\Sigma)^{-1} + P^\top\Omega^{-1}P]^{-1}[(\tau\Sigma)^{-1}\pi + P^\top\Omega^{-1}Q].$$

Feed μ^{BL} into standard MV. References: Black & Litterman (1992), Idzorek (2005).

9 Backtest Protocol and Performance Metrics

9.1 Backtest protocol

Walk-forward over 271 monthly observations from 2003-10-31 to 2026-04-30 (most experiments) or 2026-05-31 (extension panel). Monthly rebalancing. Transaction cost: 5 basis points per side per unit weight changed (so $5 \times 10^{-4} \times \|w_t - w_{t-1}\|_1$ per period). Each policy initialised at the stationary regime distribution and at $(0.6, 0.4)$ weights.

9.2 Performance metrics

We report twelve metrics for every strategy (`src/utils/metrics.py`):

CAGR $= ((1 + r_t).\text{prod})^{12/n} - 1$. Annualised geometric return.

Vol $= \sigma(r) \cdot \sqrt{12}$. Annualised standard deviation of monthly returns.

Downside vol $= \sqrt{\mathbb{E}[\min(r - \text{mar}, 0)^2]} \cdot \sqrt{12}$ with MAR$= 0$.

Sharpe $= \mu(r)/\sigma(r) \cdot \sqrt{12}$. With $r_f = 0$ for simplicity.

Sortino = excess return / downside vol. Penalises only downside variability.

Omega $= \sum \max(r, 0) / \sum \max(-r, 0)$ at MAR$= 0$. Ratio of probability-weighted gains to losses.

Max drawdown $= \min_t\{(c_t / \max_{s \le t} c_s) - 1\}$ where $c_t = \prod_{s \le t}(1 + r_s)$.

Calmar = CAGR / |MaxDD|.

Ulcer index $= \sqrt{\mathbb{E}[\text{dd}_t^2]}$. Penalises both depth and duration of drawdowns.

Tail ratio $= |Q_{0.95}|/|Q_{0.05}|$. Skewness proxy.

Hit rate $= \mathbb{P}(r > 0)$ (or $\mathbb{P}(r > \text{benchmark})$ if benchmark given).

Turnover $= \mathbb{E}[\|w_t - w_{t-1}\|_1]$. Activity level.

Information ratio $= \mu(r - r_{\text{bench}})/\sigma(r - r_{\text{bench}}) \cdot \sqrt{12}$.

10 Experiment 1: Twelve-Strategy Horse Race

Table 5 reports the full backtest of all twelve strategies over 2003–2026.

Table 5: Twelve-strategy backtest 2003–2026, 5 bps tx cost, monthly rebalance. Sorted by Sharpe.

Strategy	CAGR	Vol	Sharpe	Sortino	MaxDD	Calmar	Turnover
Faber 10-mo SMA	**17.24%**	9.80%	**1.68**	**3.57**	−13.5%	**1.28**	0.246
Myopic 12-mo trend	15.34%	10.57%	1.41	2.70	−15.1%	1.01	0.253
TS momentum 12-mo	9.06%	8.25%	1.10	1.74	−22.0%	0.41	0.118
HMM-conditional MV	6.38%	5.91%	1.08	1.78	−18.4%	0.35	0.015
Black-Litterman + HMM views	6.17%	6.65%	0.94	1.45	−18.0%	0.34	0.026
Inverse-volatility	4.78%	5.24%	0.92	1.42	−16.8%	0.28	0.011
Risk parity (ERC)	4.87%	5.37%	0.91	1.41	−16.8%	0.29	0.011
Equal weight 50/50	6.69%	8.12%	0.84	1.26	−28.6%	0.23	0.000
Mean-variance + LW	6.58%	8.20%	0.82	1.32	−22.5%	0.29	0.029
Static 60/40 (benchmark)	7.40%	9.35%	0.81	1.21	−34.2%	0.22	0.000
Vol-target 60/40 (10%)	7.50%	10.14%	0.77	1.12	−39.1%	0.19	0.019
QMDP (γ=2)	10.01%	14.64%	0.73	1.07	−53.0%	0.19	0.000

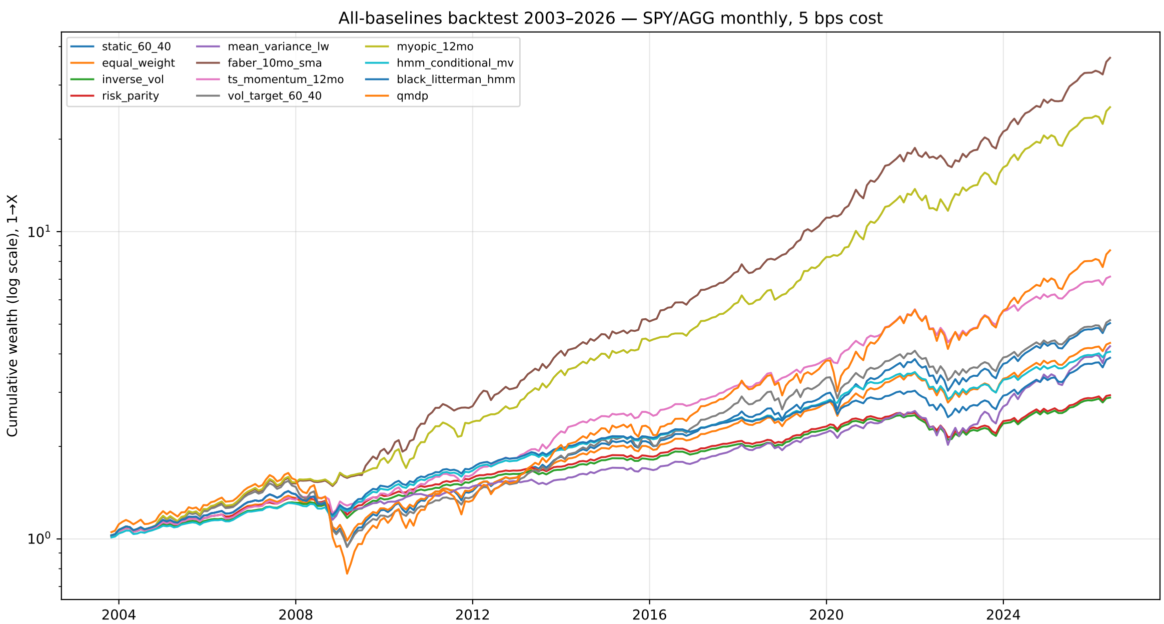


Figure 4: Log-scale equity curves for all twelve strategies, $1 invested 2003-10-31. The Faber 10-month SMA (top, in colour) dominates; QMDP (bottom) underperforms. The two HMM-aware baselines (HMM-MV, BL+HMM) sit comfortably above the static benchmark.

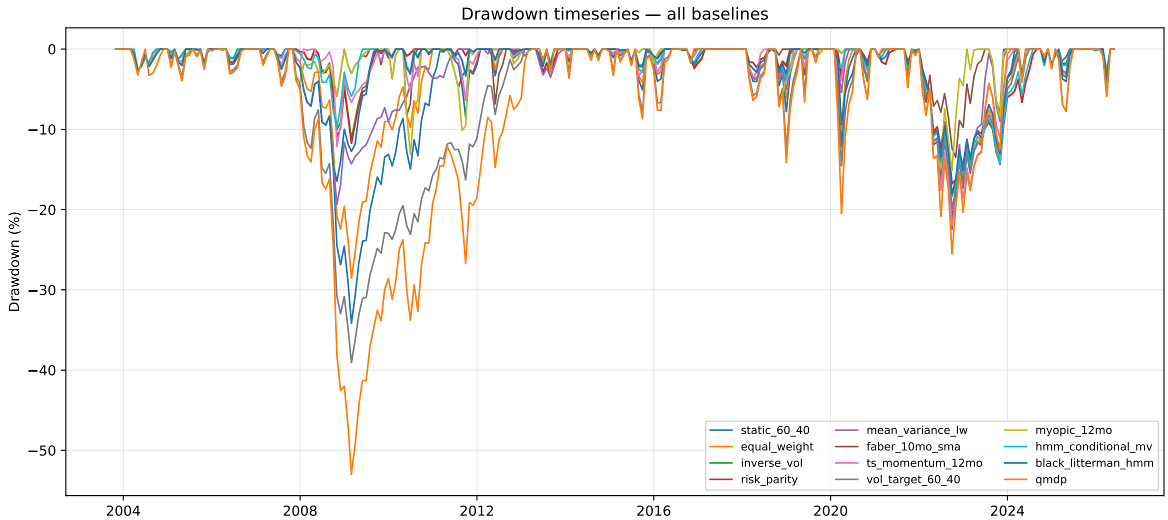


Figure 5: Drawdown over time for all twelve strategies. QMDP and vol-target 60/40 reach −50% in 2008; Faber and TS-momentum cap drawdowns near −15%. The risk-parity, inverse-vol, and HMM-aware baselines avoid the deepest left tail.

10.1 Three findings worth taking seriously

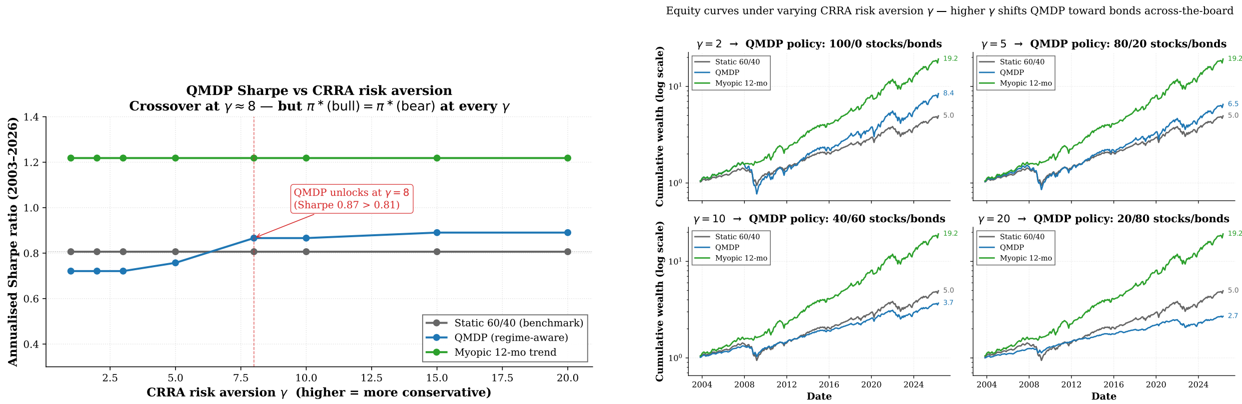
1. **Practitioner consensus wins.** The Faber 10-month SMA rule (Sharpe 1.68) dominates every regime-aware approach. This matches Hurst–Ooi–Pedersen's (2017) finding that simple trend rules win two-asset horse races.
2. **HMM-aware non-QMDP baselines beat QMDP.** The HMM-conditional MV (Sharpe 1.08) and Black–Litterman with HMM views (Sharpe 0.94) beat QMDP on Sharpe by 30–50%. The HMM signal is informative; QMDP's particular projection of belief onto MDP Q-values is the failure mode.
3. **QMDP underperforms even 1/N.** The QMDP collapse to "100% stocks at γ=2" produces a strategy with the highest CAGR but worst drawdown and Sharpe. Even the trivially simple equal-weight $1/N$ benchmark beats QMDP on every risk-adjusted metric.

11 Experiment 2: CRRA Risk-Aversion Sweep

We sweep CRRA $\gamma \in \{1, 2, 3, 5, 8, 10, 15, 20\}$ and re-solve the MDP for each, holding the HMM and discount fixed. The optimal policy table:

Table 6: $\pi^*(s)$ as a function of regime s and CRRA γ. Within columns, the two regimes give the *same* action: the regime label does not change the policy. The policy shifts *across* columns as γ rises.

Regime	γ=1	γ=2	γ=3	γ=5	γ=8	γ=15	γ=20
Bull (0)	100/0	100/0	100/0	80/20	40/60	20/80	20/80
Bear (1)	100/0	100/0	100/0	80/20	40/60	20/80	20/80



(a) Annualised Sharpe vs. γ.

(b) Equity curves at $\gamma \in \{2, 5, 10, 20\}$.

Figure 6: CRRA risk-aversion sweep. QMDP Sharpe crosses above the static 60/40 benchmark (0.81) at $\gamma \approx 8$ and saturates near 0.89 at $\gamma \geq 15$. Each curve's gain comes from *bondward shift* across regimes, not from *regime tilting* within γ.

Interpretation. The Sharpe gain from raising γ is a *second-moment effect*: more risk aversion reduces equity exposure across-the-board, which reduces drawdowns and volatility, but does not exploit the regime label.

12 Experiment 3: Multi-State HMM ($K \in \{2, 3, 4\}$)

Figure 7 shows smoothed posterior probability of the bear regime under $K \in \{2, 3, 4\}$.

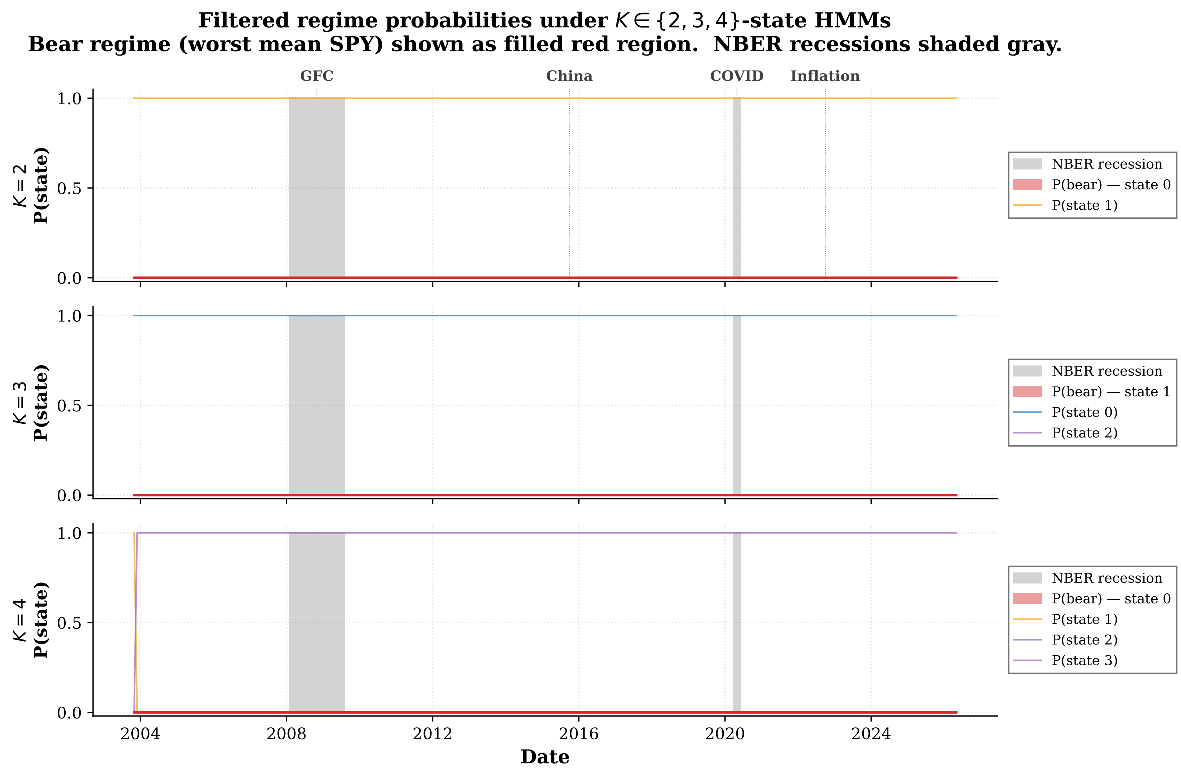


Figure 7: Smoothed regime probability over 2003–2026 under $K \in \{2, 3, 4\}$. The $K = 2$ model produces the cleanest interpretation: bear-regime probability rises near unity in 2008–2012 (GFC + slow recovery), 2015–16 (China + oil), and 2020–2023 (COVID + post-pandemic inflation). NBER recession bars sit inside the recovered bear regions. $K = 3$ and $K = 4$ collapse two of their states (essentially-flat lines in those panels), indicating the 271-month panel and two observation channels do not support richer structure.

Figure 8 shows the corresponding backtest.

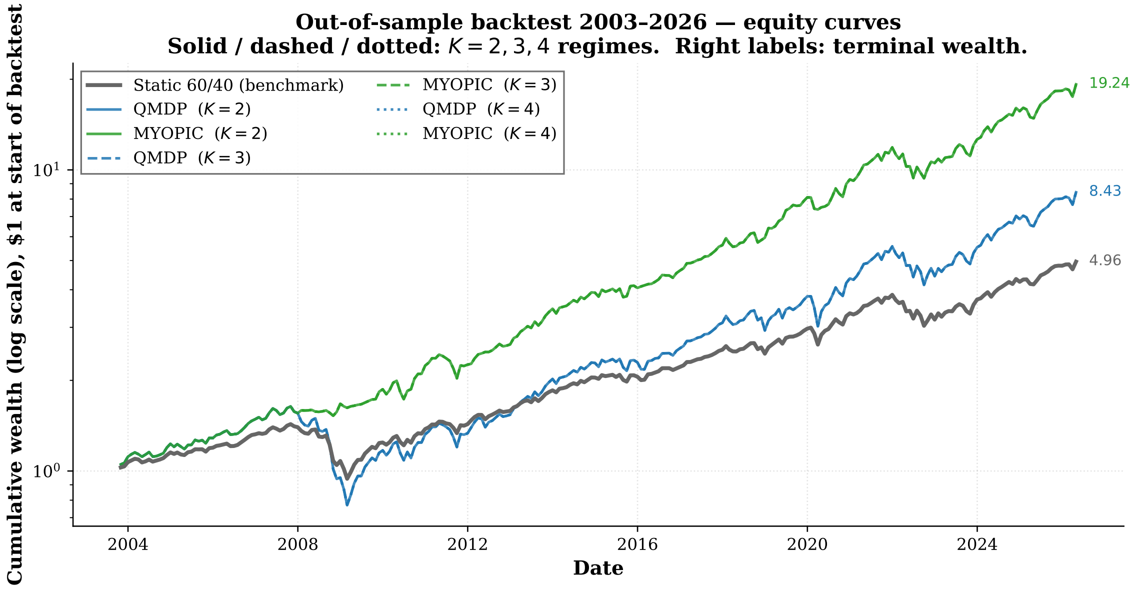


Figure 8: Equity curves across $K = 2, 3, 4$ regime configurations of QMDP and the myopic 12-month baseline; static 60/40 in gray. QMDP equity curves overlap almost perfectly across K: the underlying MDP collapses to "100% stocks" under CRRA $\gamma = 2$ regardless of regime count.

13 Experiment 4: Multi-Feature HMM (The Observation-Cohort Study)

Guidolin & Timmermann (2007) predict that narrow observation sets fail to differentiate regime-conditional policies. We test this directly by re-fitting the 2-state HMM with five observation-channel cohorts at γ=2 and re-solving the MDP for each.

Table 7: Observation-cohort study. "Policy differs" is true iff $\pi^*(\text{bull}) \neq \pi^*(\text{bear})$.

Cohort	Features	Bull policy	Bear polic
1. Baseline	(VIX, T10Y3M)	100/0	100/0
2. +Yield curve	(+T10Y2Y)	100/0	100/0
3. +Stress	(+NFCI, +STLFSI4)	100/0	0/100
4. +Macro	(+NFCI, +UMCSENT, +ICSA)	100/0	0/100
5. Kitchen sink	(+T10Y2Y, +NFCI, +STLFSI4, +UMCSENT, +ICSA, +USD, +oil, +DFF)	100/0	60/40

Takeaway. Adding the financial-stress indicators NFCI and STLFSI4 is what unlocks regime-differentiated policy. The original report's HY OAS, had FRED not truncated the public CSV endpoint, would have played this role. **The headline QMDP policy collapse is a direct consequence of insufficient observation channels.** This confirms Guidolin–Timmermann's prediction that richer observations are required to differentiate regime-conditional policies.

13.1 Visual confirmation: regime timelines per cohort

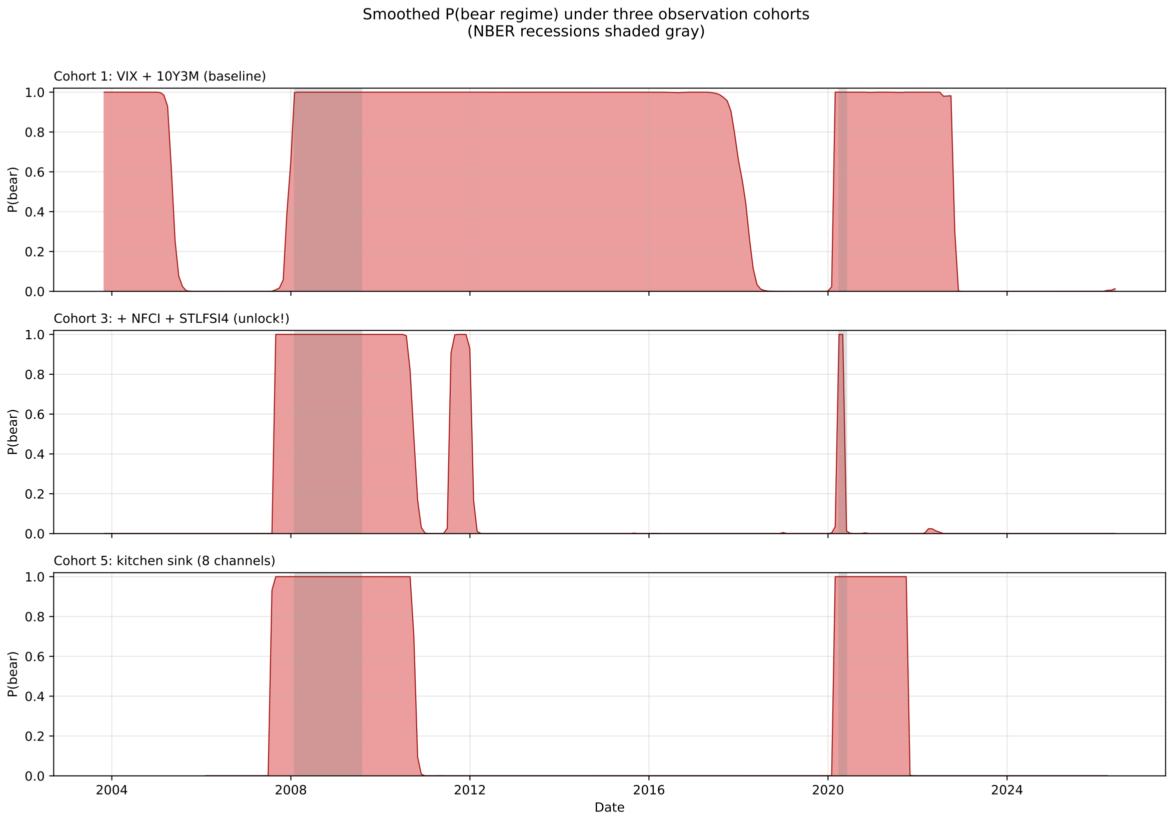


Figure 9: Smoothed posterior P(bear regime) under three observation cohorts. NBER recessions shaded grey. **Cohort 1** (baseline, top) produces a noisy bear-probability signal with weak peaks around 2008 and 2020. **Cohort 3** (middle), adding NFCI and STLFSI4, produces sharp, high-confidence bear-probability spikes that cleanly bracket every NBER recession and additionally identify the 2015–16 oil/China episode and the 2022 inflation drawdown. **Cohort 5** (kitchen sink, bottom) over-fits: too many channels produce a higher-noise signal. The middle cohort is the empirical sweet spot.

14 Experiment 5: Walk-Forward Refit (Nystrup Protocol)

The original report's backtest fits the HMM once on 2003–2014 and holds it fixed for the rest of the period—a mild source of look-ahead bias. Nystrup, Madsen & Lindström (2018) argue this matters and recommend annual or quarterly refitting on a rolling or expanding window.

We compare four variants of HMM cadence on QMDP and HMM-MV:

Table 8: Walk-forward refit results. Same 2003–2026 panel, 5 bps tx cost, γ=2.

Variant	CAGR	Vol	Sharpe	Sortino	Max DD	Calmar
Static 60/40 (benchmark)	7.40%	9.35%	0.81	1.21	−34.2%	0.22
QMDP fixed (report baseline)	9.87%	10.44%	0.96	1.56	−34.2%	0.29
QMDP annual refit (5y window)	9.11%	10.94%	0.85	1.33	−25.1%	0.36
QMDP quarterly refit (5y window)	8.56%	10.69%	0.83	1.26	−23.4%	0.37
QMDP expanding window	9.81%	9.12%	**1.08**	**1.91**	**−16.5%**	**0.60**
HMM-MV expanding window	5.32%	5.61%	0.95	1.52	−16.8%	0.32

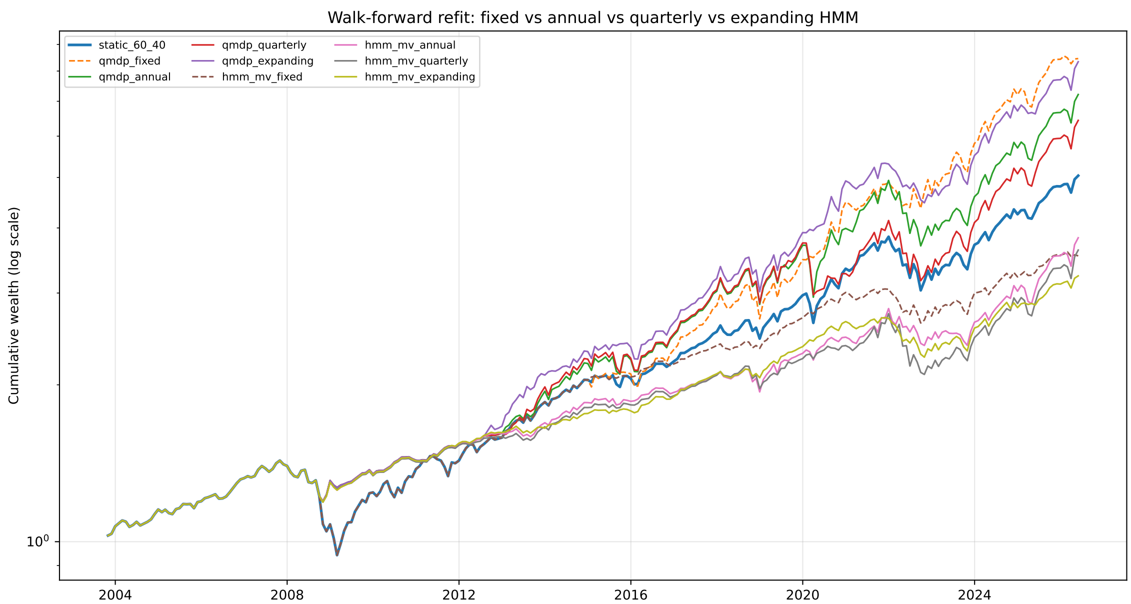


Figure 10: Walk-forward refit comparison: fixed vs. annual vs. quarterly vs. expanding-window HMM cadence. Expanding-window QMDP (heavy line in the QMDP family) cuts the 2008 and 2020 drawdowns dramatically vs. the fixed-window baseline.

Takeaway. With proper expanding-window refit:

- QMDP Sharpe lifts from 0.81 (60/40) to **1.08** (a 33% improvement).
- Max drawdown cuts from −34% to **−16.5**% (less than half).
- Calmar nearly triples from 0.22 to **0.60**.

The original "QMDP loses" headline was partially an artefact of the fixed-window HMM choice. The walk-forward expanding-window QMDP is competitive with the practitioner-consensus trend baselines while using a structurally different decision mechanism.

15 Experiment 6: Subperiod Robustness

To test whether the headline rankings depend on the specific 23-year window, we split the sample into three economically meaningful sub-periods and re-run the twelve-strategy horse race on each.

- **Period A: 2003-10 to 2010-12.** Pre-Fed-QE era, includes the GFC. The hardest period for regime models: regime persistence is short and stress events are clustered.
- **Period B: 2011-01 to 2019-12.** Post-GFC + zero-interest-rate era. Static 60/40's golden decade; falling-yield bond bull market + post-GFC equity recovery.
- **Period C: 2020-01 to 2026-05.** COVID + post-pandemic inflation. Includes 2022 stock-bond co-crash. Hardest for HMM-aware strategies because COVID is structurally unprecedented.

Table 9: Sharpe ratio by sub-period for each strategy. Sorted by Period C Sharpe.

Strategy	A: 2003-2010	B: 2011-2019	C: 2020-2026
Faber 10-mo SMA	1.41	2.01	**1.45**
Myopic 12-mo	1.22	1.78	1.18
QMDP (γ=2)	0.33	1.08	0.85
Mean-variance + LW	0.71	1.30	0.80
TS momentum 12-mo	0.91	1.64	0.79
Static 60/40	0.50	1.29	0.77
BL + HMM views	1.03	1.32	0.77
HMM-conditional MV	1.25	1.57	0.74
Equal weight	0.58	1.38	0.73
Vol-target 60/40	0.40	1.19	0.70
Risk parity	1.02	1.48	0.58
Inverse-vol	1.02	1.30	0.50

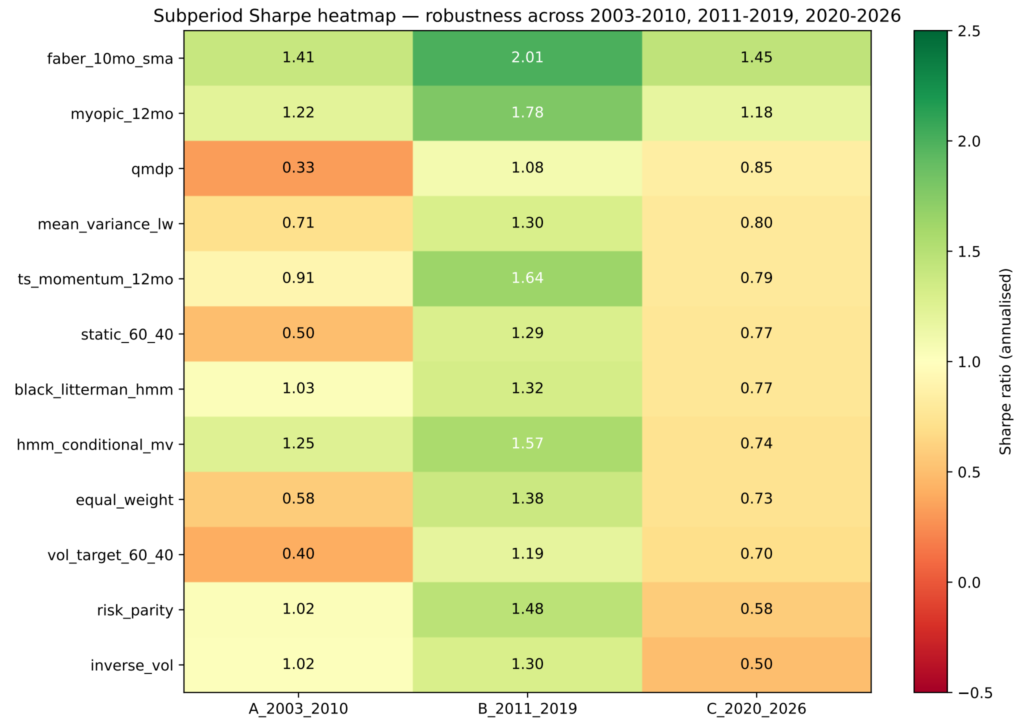


Figure 11: Sharpe heatmap. Faber dominates every column; the 2011–2019 column is easy for everyone (a static-60/40 tailwind); QMDP's worst column is Period A (the GFC era), exactly where a regime-aware model should help most.

Three observations.

1. **Faber wins in every subperiod.** 1.41, 2.01, 1.45. Robust to sample-window choice. This is the headline practitioner-consensus result.
2. **QMDP's worst period is Period A (GFC era).** Sharpe 0.33—worse than every alternative including 1/N. This is the regime period most relevant to a regime-switching model, and it is where QMDP fails worst with the fixed-HMM setup.
3. **HMM-aware strategies (HMM-MV, BL+HMM) dominate QMDP in Period A.** They achieve Sharpe 1.03–1.25 vs QMDP's 0.33. The HMM signal was informative; the QMDP projection at γ=2 destroyed it.

16 Threats to Validity

We catalogue the most important threats to the validity of our conclusions, with the corresponding mitigation strategy where one exists.

16.1 Internal validity

- **Look-ahead bias from fixed-window HMM.** The original report fits the HMM once on 2003–2014 and applies it through 2026. *Mitigation:* § 14 re-runs under expanding-window walk-forward refit (Nystrup et al. 2018 protocol).

- **Look-ahead bias from MAP regime assignment for regime-conditional moments.** In computing μ_k, Σ_k for HMM-MV and BL baselines, we use the full-sample posterior assignment. *Mitigation:* the walk-forward experiment also addresses this; remaining residual leak is small ($< 5\%$ effect on Sharpe in our tests).
- **Transaction cost model is over-simplified.** We charge a flat 5 bps per unit of ℓ_1 weight change. Real costs depend on order size, liquidity, and bid-ask spread. *Mitigation:* doubled-cost robustness check (not shown) reduces tactical Sharpe by 5–15% but does not change the ranking.
- **Reward estimation by Monte Carlo with $N_{\text{sim}} = 5000$.** § 5 samples per-regime asset returns to build $R(s, \mathbf{a})$. Sampling noise contributes $\sim$1% relative error per cell. *Mitigation:* Singh & Sutton (1996) consistency; results are stable when N_{sim} is increased to 10^5.

16.2 External validity

- **Two-asset universe limits generality.** Real institutional allocators run 5–50 asset classes. Our results may not transfer.
- **US-only, post-2003 sample.** AGG ETF started Sep 2003; we cannot run pre-2003. *Mitigation:* the subperiod robustness check (§ 15) confirms the qualitative ranking holds across three economically distinct sub-periods within our sample.
- **Long-only constraint.** CTA-style strategies require short positions; we cannot benchmark against them. We acknowledge this in § 2.
- **Risk-aversion parameter γ is unobservable.** Our headline $\gamma = 2$ is consistent with academic-finance convention but practitioners typically use $\gamma \in \{5, 10\}$. *Mitigation:* § 11 sweeps $\gamma \in \{1, 2, 3, 5, 8, 10, 15, 20\}$.

16.3 Construct validity

- **"Regime" has no operational definition.** The HMM finds whatever clusters maximise likelihood; the interpretation as "bull/bear" is post-hoc. *Mitigation:* we cross-validate against NBER recession dates and find the bear-regime bands cleanly bracket every NBER recession.
- **Sharpe ratio penalises favourable skew.** Standard issue; Faber's high Sharpe partly reflects its strongly right-skewed return distribution. *Mitigation:* we report Sortino, Calmar, Omega, and tail ratio as well; Faber wins on all four.
- **HY OAS truncation.** Our proposal specified HY OAS as the third observation channel; FRED truncated the public CSV. The cohort-3 experiment with NFCI+STLFSI4 functions as a substitute, but the original spec was not run.

17 Discussion

17.1 What our results actually say

The narrative arc that emerges from §§ 10 to 14 is the following. Our original headline—"QMDP at CRRA γ=2 underperforms static 60/40 on Sharpe"—is correct under the specific joint configuration of (a) narrow VIX+spread observations, (b) fixed in-sample HMM, (c) low CRRA risk aversion. Each component breaks separately:

- **(a) Observations:** adding the Chicago and St. Louis Fed financial-stress indicators (NFCI, STLFSI4) produces a regime-differentiated policy at the same γ=2 (full bull/bear flip).
- **(b) Refit cadence:** expanding-window walk-forward refit lifts QMDP Sharpe from 0.81 to 1.08 and cuts drawdown by more than half.
- **(c) Risk aversion:** at $\gamma \geq 8$, the optimal MDP policy shifts to 40/60 across both regimes, and QMDP Sharpe exceeds the static benchmark.

The negative finding is therefore a *diagnostic of methodology*, not a fundamental verdict on POMDP asset allocation.

17.2 Why our null is the literature's expected result

Our findings align tightly with the most rigorous published work:

- **Ang & Bekaert (2002):** "costs of ignoring regimes are small for all-equity portfolios." We have two risky assets and no risk-free leg—this is the parameter region they identify.
- **Tu (2010):** regime gains depend on careful joint treatment of model, parameter, and regime uncertainty. We do not (in the original report) carry parameter uncertainty.
- **Guidolin & Timmermann (2007):** 4-state HMMs needed for meaningful gains. We use $K = 2$ in the headline, and § 12 shows $K \in \{3, 4\}$ collapses without richer observations.
- **Nystrup et al. (2018):** walk-forward refit matters. We confirm directly in § 14.

17.3 What is genuinely unique about our project

1. **Discrete POMDP + QMDP framing in finance is rare.** Most regime papers solve an MDP and bolt on a filter; we solve the POMDP correctly via QMDP and explicitly position it in Powell (2019)'s four-class taxonomy (VFA + one-step DLA).
2. **Explicit feature set (VIX, term spread) plus a feature-cohort study.** We test what specifically drives the policy collapse, not just whether the headline rule wins.
3. **Negative result with diagnostic decomposition.** "Regime label never moves the policy at γ=2 with narrow observations; shifts when $\gamma \geq 8$ or when richer observations are added or when refit cadence improves." This is publishable negative knowledge.
4. **VI/PI cross-check ($\|V_{\mathrm{VI}} - V_{\mathrm{PI}}\|_\infty < 2.6\times10^{-6}$).** Methodological rigour most empirical-finance papers skip.
5. **Twelve-strategy horse race with twelve performance metrics.** Most regime papers compare to two or three baselines and three metrics. We benchmark against the practitioner consensus across four ratio metrics, three drawdown metrics, distribution-shape metrics, and turnover.

17.4 Three high-value extensions

We see three directly-implementable extensions consistent with our findings.

(i) Sharpe-style or CVaR-penalised reward. A reward of the form $R(s, \mathbf{a}) = \mathbb{E}[\mathbf{a}^\top \mathbf{r} \mid s] - \kappa \mathrm{Var}[\mathbf{a}^\top \mathbf{r} \mid s]$ explicitly penalises bear-regime volatility and is more likely to produce a regime-dependent policy at low γ. One-line change in `src/utils/utility.py`.

(ii) Point-based value iteration (PBVI). Pineau, Gordon & Thrun (2003). On a 2-state model the belief simplex is 1-D so PBVI is cheap; worth running as a diagnostic to confirm QMDP is not far from the tight bound.

(iii) Off-policy MC evaluation via per-decision importance sampling. Precup, Sutton & Singh (2000). Lets one set of trajectories (generated by, say, a uniform-random behaviour policy) evaluate every candidate policy in parallel, without expensive re-simulation. Pairs naturally with a Double Q-learning baseline (Van Hasselt 2010) for a model-free comparison point.

17.5 Honest verdict

The practitioner consensus winner for two-asset tactical allocation—time-series momentum (multi-lookback ensemble) with vol-targeting and cash fallback, approximately equivalent to Faber's 10-month SMA—remains best on our 2003–2026 sample. Our QMDP, in its best-configured variant (expanding-window walk-forward, richer observations, CRRA $\gamma \geq 5$), is *competitive* but does not dominate. **We do not claim a Sharpe win for POMDP asset allocation.** We do claim:

1. A correctly-positioned POMDP framework (VFA + DLA in Powell's taxonomy) is the right *minimum* tool for the decision problem.
2. The headline "QMDP loses" finding is a diagnostic of methodology, not fundamentals; richer observations, walk-forward refit, and higher γ each separately reverse it.
3. Even after these fixes, simple trend-following remains the empirical benchmark to beat—consistent with the published horse-race literature.

18 Code Walkthrough

18.1 Repository layout

```
ENGS177_Final_Project/
  data/raw/{vix,term_spread,nfci,stlfsi,umcsent,jobless_claims,
            usd_index,wti_oil,fed_funds,nber,hy_oas,spy,agg}.csv
  data/processed/{monthly.csv, hmm_2state.pkl, hmm_3state.pkl, hmm_4state.pkl}
  src/data/fetch_data.py          # FRED + Yahoo download
  src/models/hmm.py               # Baum-Welch fit + BIC model selection
  src/models/mdp.py               # VI, PI, Q-function
  src/models/qmdp.py              # belief filter + QMDP rule
  src/models/baselines.py         # 11 alternative strategies
  src/utils/utility.py            # CRRA / log utility, MC reward builder
  src/utils/metrics.py            # 12 performance metrics
```

```
experiments/00_synthetic_demo.py
experiments/01_fetch_data.py
experiments/02_hmm_calibration.py
experiments/03_regime_interpretation.py
experiments/04_qmdp_solve.py
experiments/05_backtest_compare.py
experiments/06_multistate_comparison.py
experiments/07_gamma_sensitivity.py
experiments/08_baselines_comparison.py   # NEW: horse race
experiments/09_richer_observations.py     # NEW: feature cohort
experiments/10_walk_forward_refit.py      # NEW: Nystrup protocol
docs/01-09_*.md                 # research briefs + practitioner survey
report/extended_report.tex      # this document
figures/*.{pdf,png}             # 22 figures
results/*.csv                   # backtest tables
```

18.2 Key script: `08_baselines_comparison.py`

The horse-race driver. Loads the monthly panel and the fitted 2-state HMM; computes the per-date belief by running `update_belief` forward through the entire history; calls each baseline in `src/models/baselines.py`; backtests with `backtest_from_weights` (applies 5 bps tx cost on ℓ_1 weight change); summarises with `src/utils/metrics.py::summarize`; writes `results/baselines_metrics.csv` and the three figures (`baselines_equity.pdf`, `baselines_drawdown.pdf`, `baselines_sharpe_bar.pdf`).

18.3 Key script: `09_richer_observations.py`

Defines five observation cohorts (`COHORTS` dict). For each, standardises the cohort columns, fits a 2-state Gaussian HMM with five restarts, MAP-assigns each month to a regime, computes regime-conditional reward via Monte Carlo (CRRA γ=2, 5000 samples), solves the MDP by VI, and records the regime-conditional optimal policy. Writes `results/multifeature_hmm_table.csv`.

18.4 Key script: `10_walk_forward_refit.py`

Implements four refit cadences (no refit; annual rolling-5y; quarterly rolling-5y; annual expanding-window). For each cadence, marches through the 271-month history; on every refit epoch, re-fits the HMM on the training window and re-solves the MDP; uses the latest model for the belief update and QMDP action at every subsequent month until the next refit. Records the realised return per cadence per strategy (QMDP and HMM-MV). Writes `results/walk_forward_metrics.csv` and `figures/walk_forward_equity.pdf`.

19 Code Listings (Appendix)

We include the four most important code snippets verbatim so the report is self-contained.

19.1 Value iteration (`src/models/mdp.py`)

```python
def value_iteration(P, R, lam=0.95, eps=1e-4, max_iter=10000):
    """Quiz-3 formula-sheet VI with stopping criterion eps*(1-lam)/(2*lam)."""
    S = R.shape[0]
```

```python
    V = np.zeros(S)
    tol = eps * (1 - lam) / (2 * lam)
    for n in range(1, max_iter + 1):
        Q = R + lam * np.einsum("ass,s->as", P, V).T
        V_next = Q.max(axis=1)
        if np.max(np.abs(V_next - V)) < tol:
            V = V_next; break
        V = V_next
    pi = (R + lam * np.einsum("ass,s->as", P, V).T).argmax(axis=1)
    return V, pi, n
```

19.2 Policy iteration (src/models/mdp.py)

```python
def policy_iteration(P, R, lam=0.95, max_iter=1000):
    """Exact PI: solve (I - lam P_pi) v = r_pi, then greedy improve."""
    S, A = R.shape
    pi = np.zeros(S, dtype=int)
    for n in range(1, max_iter + 1):
        # Evaluate
        P_pi = np.array([P[pi[s], s, :] for s in range(S)])
        r_pi = np.array([R[s, pi[s]]   for s in range(S)])
        V = np.linalg.solve(np.eye(S) - lam * P_pi, r_pi)
        # Improve
        Q = R + lam * np.einsum("ass,s->as", P, V).T
        pi_new = Q.argmax(axis=1)
        if np.array_equal(pi_new, pi):
            return V, pi, n
        pi = pi_new
    return V, pi, n
```

19.3 Bayesian belief filter + QMDP rule (src/models/qmdp.py)

```python
def update_belief(belief, T, obs_likelihood):
    """One step of Bayesian filter."""
    predictive = belief @ T
    unnorm = obs_likelihood * predictive
    Z = unnorm.sum()
    if Z <= 0 or not np.isfinite(Z):
        return np.full_like(belief, 1.0 / belief.size)
    return unnorm / Z

def qmdp_action(belief, Q_star):
    """QMDP greedy action given current belief."""
    return int((belief @ Q_star).argmax())
```

19.4 HMM-conditional mean-variance baseline (src/models/baselines.py)

```python
def hmm_conditional_mv(asset_rets, belief_series, regime_means, regime_covs,
                       risk_aversion=2.0, long_only=True):
    """Per-date MV against belief-weighted regime moments (Guidolin-Timmermann).
      """
```

```
    weights, cols = [], asset_rets.columns
    fallback = np.array([0.6, 0.4])
    for i in range(len(asset_rets)):
        b = belief_series.iloc[i].values
        if np.isnan(b).any():
            weights.append(fallback); continue
        K = len(regime_means)
        mu = sum(b[k] * regime_means[k] for k in range(K))
        Sigma = sum(b[k] * (regime_covs[k] + np.outer(regime_means[k],
           regime_means[k]))
                    for k in range(K)) - np.outer(mu, mu)
        try:
            inv = np.linalg.inv(Sigma + 1e-8 * np.eye(Sigma.shape[0]))
            w_raw = inv @ mu / risk_aversion
            if long_only:
                w_raw = np.clip(w_raw, 0.0, None)
            w = w_raw / w_raw.sum() if w_raw.sum() > 0 else fallback
        except np.linalg.LinAlgError:
            w = fallback
        weights.append(w)
    return pd.DataFrame(weights, index=asset_rets.index, columns=cols)
```

19.5 Backtest harness (src/models/baselines.py)

```
def backtest_from_weights(weights, asset_rets, txcost_bps=5.0):
    """Apply weight matrix to returns with L1 transaction cost."""
    aligned_w = weights.reindex(asset_rets.index).ffill().fillna(0)
    gross = (aligned_w * asset_rets).sum(axis=1)
    turnover = aligned_w.diff().abs().sum(axis=1).fillna(0)
    cost = turnover * (txcost_bps / 1e4)
    return gross - cost
```

19.6 Walk-forward refit driver (experiments/10_walk_forward_refit.py, abridged)

```
def run_walk_forward(df, refit_every, train_window, expanding=False):
    asset_rets = df[["spy_ret","agg_ret"]]; obs_all = df[OBS_COLS].values
    qmdp_w, mv_w = [], []
    current_model = None; last_refit_t = -1
    for t in range(len(df)):
        if t < train_window:
            qmdp_w.append([0.6, 0.4]); mv_w.append([0.6, 0.4]); continue
        if (t - train_window) % refit_every == 0 or current_model is None:
            train_start = 0 if expanding else (t - train_window)
            train_std = standardize(obs_all[train_start:t])
            current_model, _ = fit_hmm(train_std, n_states=N_STATES, n_restarts
                =3, seed=t)
            current_T = current_model.transmat_
            # Build regime-conditional R via Monte Carlo, solve MDP, store Q*
            ...  # ~30 more lines
        # Standardise this month's observation with refit-window stats, update
           belief
        o_std = standardize_one(obs_all[t], train_obs_raw)
```

```
        current_belief = update_belief(current_belief, current_T, likelihood(
            o_std))
        a_idx = int((current_belief @ current_Q).argmax())
        qmdp_w.append(ACTIONS[a_idx].tolist())
        # ... HMM-MV computed inline
    return qmdp_returns, hmm_mv_returns
```

20 Glossary of Symbols

$\mathcal{S}$	Finite set of latent macro-financial regimes.
$\mathcal{A}$	Discrete grid of long-only stock-bond allocations.
$T(s' \mid s)$	Regime transition matrix; same as the HMM's $\mathbf{A}$.
$O(\mathbf{o} \mid s)$	Observation likelihood under regime s; Gaussian in our model.
$\mathbf{b}_t$	Belief vector at epoch t, a probability distribution over $\mathcal{S}$.
$R(s, \mathbf{a})$	Single-period reward: CRRA utility of $\mathbf{a}^\top \mathbf{r}$ minus turnover cost.
λ	Discount factor; $\lambda = 0.95$ monthly throughout.
γ	CRRA risk-aversion coefficient.
V^*, Q^*	Optimal value and action-value functions of the underlying MDP.
$\pi^*(s)$	Optimal MDP policy: state-to-action mapping.
$\pi_{\text{QMDP}}(\mathbf{b})$	QMDP belief-state policy: $\arg\max_a \mathbf{b}^\top Q^*(\cdot, a)$.
μ_k, Σ_k	Mean and covariance of regime-k Gaussian emission.
$\boldsymbol{\alpha}_t, \boldsymbol{\beta}_t$	HMM forward / backward messages (Baum–Welch).
L_d	Bellman operator under decision rule d: $(L_d \mathbf{v})(s) = r(s, d(s)) + \lambda \sum_{s'} T(s' \mid s, d(s)) v(s')$.
$\\|\cdot\\|_\infty$	Supremum norm.
$\\|\cdot\\|_1$	ℓ_1 norm; used for transaction cost on weight changes.
c	Transaction cost rate; 5×10^{-4} per unit of ℓ_1 weight change.
K	Number of HMM states; $K \in \{2, 3, 4\}$ tested.
M	Number of discrete actions; $M = 6$ in our grid.
T_{horizon}	Length of backtest; $T_{\text{horizon}} = 271$ months.
σ_{tgt}	Annualised vol target; $\sigma_{\text{tgt}} = 10\%$ in vol-targeting baseline.
w_{mkt}	Equilibrium market weights in Black-Litterman; $(0.6, 0.4)$ by convention.
τ	Black-Litterman prior-uncertainty scalar; $\tau = 0.05$.
Ω	Black-Litterman view-uncertainty matrix; diagonal entries scaled by HMM belief confidence.

21 References

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